

Date: 15-01-25

## Graphs (3D)

Day: MTWTFSS

①  $f(x, y) = 6 - 3x - 2y$ .

Solution:-

Graph:  $z = f(x, y)$

$$z = 6 - 3x - 2y.$$

②  $g(x, y) = \sqrt{9 - x^2 - y^2}$ .

Solution:-

Graph:  $z = g(x, y)$

$$z = \sqrt{9 - x^2 - y^2}$$

③  $h(x, y) = 4x^2 + y^2$

Solution:-

Graph:  $z = h(x, y)$

$$z = 4x^2 + y^2.$$

④  $f(x, y) = (x^2 + 3y^2) e^{-x^2 - y^2}$

Solution:-

Graph:  $z = f(x, y)$

$$z = (x^2 + 3y^2) e^{-(x^2 + y^2)}$$

⑤  $f(x, y) = \sin x + \sin y$ .

Solution:-

Graph:  $z = f(x, y)$

$$z = \sin x + \sin y$$

⑥  $f(x, y) = \sin x + \cos y$

Solution:-

Graph:  $z = f(x, y)$

$$z = \sin x + \cos y.$$

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Functions of three or more variables :-

$$\textcircled{1} \quad h = f(u, v, w) = 4u^3 + 3v^2 + w.$$

$$\textcircled{2} \quad g = f(u, v, w, x) = 5u^2 + x + 6x^2 + 5w + \sqrt{v}$$

$$\star \quad p = f(x, y, z) = \ln(z-y) + xy \sin z$$

$$\star \quad h = f(x, y, z) = x^2 + y^2 + z^2$$

$$z - y > 0$$

$$z > 0$$

$$z = 0$$